Double-click (or enter) to edit

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```
from google.colab import files
files.upload()
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
data = pd.read_csv("Pune Restaurants.csv")
data.head()
```

	Restaurant_Name	Category	Pricing_for_2	Locality	Dining_Rating	Dining_Review_Count	Delivery_Rating	Delivery_Rating_Count	
0	Santè Spa Cuisine	Continental, Healthy Food, Mediterranean	1200	Koregaon Park, Pune	4.9	1469	3.9	588	http:
1	Le Plaisir	Cafe, Italian, Continental, Salad, Sandwich, P...	1000	Deccan Gymkhana, Pune	4.9	4808	4.3	4959	I
2	Gong	Chinese, Sushi, Asian, Momos, Beverages	1700	Balewadi High Street, Baner, Pune	4.9	1788	4.3	1352	http
3	The French Window Patisserie	Cafe, Desserts, French, Bakery, European	600	Koregaon Park, Pune	4.9	1643	4.4	1208	ht
4	Savya Rasa	South Indian, Mangalorean, Kerala, Chettinad, ...	2100	Koregaon Park, Pune	4.9	1283	4.3	446	https

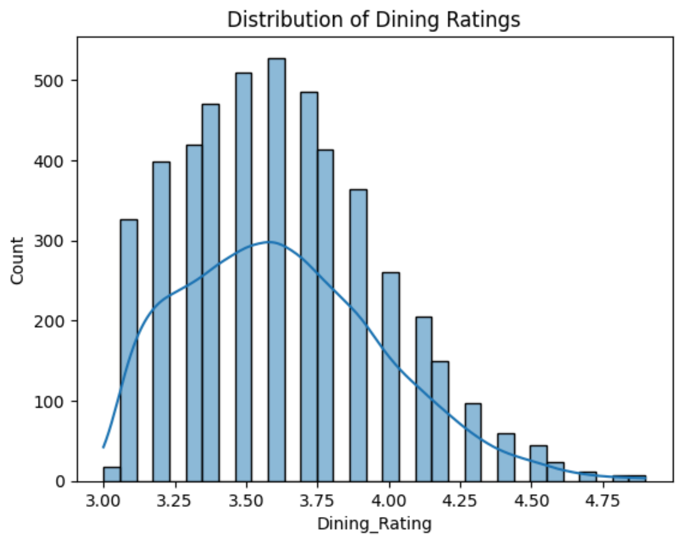
Next steps: Generate code with data New interactive sheet

```
data.shape
data.info()
data.describe()
```

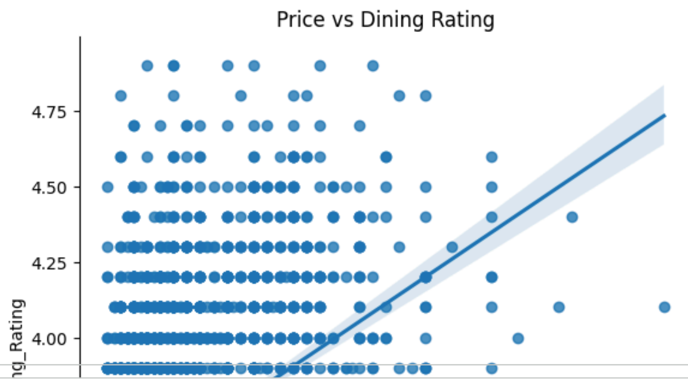
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4797 entries, 0 to 4796
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Restaurant_Name       4797 non-null   object 
 1   Category              4797 non-null   object 
 2   Pricing_for_2         4797 non-null   int64  
 3   Locality              4797 non-null   object 
 4   Dining_Rating         4797 non-null   float64
 5   Dining_Review_Count   4797 non-null   int64  
 6   Delivery_Rating       3226 non-null   float64
 7   Delivery_Rating_Count 4797 non-null   int64  
 8   Website               4797 non-null   object 
 9   Address               4797 non-null   object 
 10  Phone_No              4797 non-null   object 
 11  Latitude              4797 non-null   float64
 12  Longitude             4797 non-null   float64
 13  Known_for1            4155 non-null   object 
 14  Known_for2            1078 non-null   object 
dtypes: float64(4), int64(3), object(8)
memory usage: 562.3+ KB
```

	Pricing_for_2	Dining_Rating	Dining_Review_Count	Delivery_Rating	Delivery_Rating_Count	Latitude	Longitude
count	4797.000000	4797.000000	4797.000000	3226.000000	4797.000000	4797.000000	4797.000000
mean	571.867834	3.629748	182.439233	3.837260	913.103189	18.549262	73.835703
std	425.309842	0.350933	482.759166	0.284843	2522.973728	0.067471	0.085912
min	100.000000	3.000000	0.000000	2.400000	0.000000	18.321548	73.297426
25%	300.000000	3.400000	11.000000	3.700000	7.000000	18.503244	73.790063
50%	450.000000	3.600000	36.000000	3.900000	82.000000	18.540623	73.837188
75%	700.000000	3.900000	137.000000	4.000000	605.000000	18.590621	73.897032
max	4300.000000	4.900000	8152.000000	4.500000	56700.000000	18.842622	74.177275

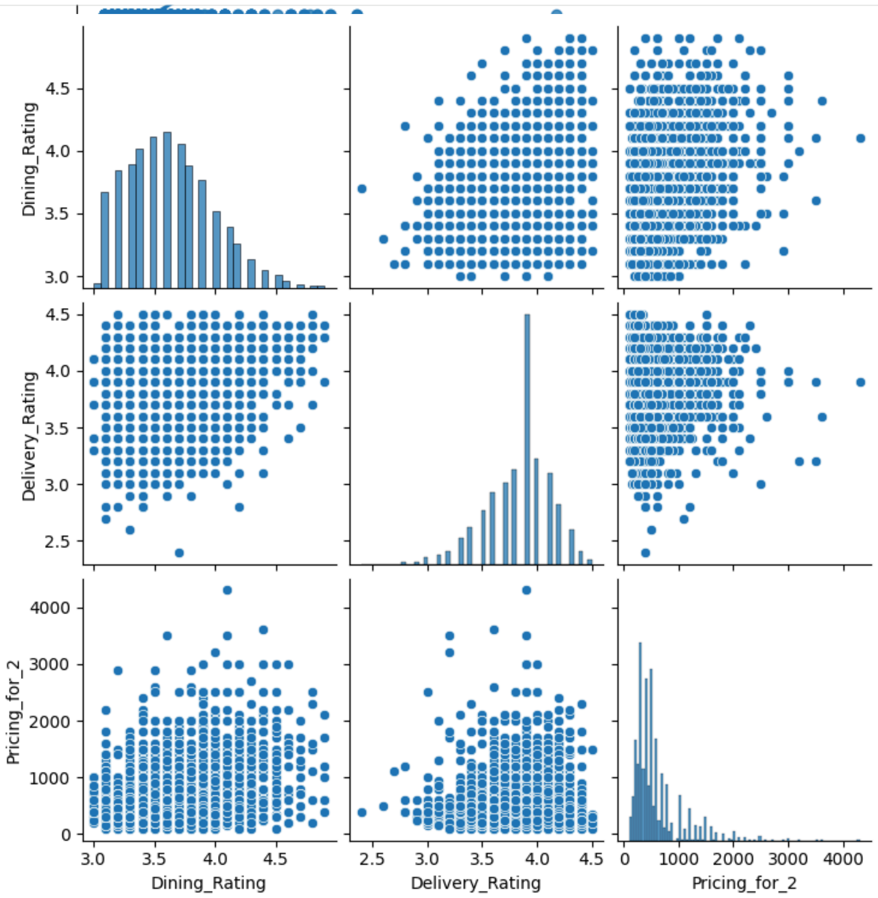
```
sns.histplot(data['Dining_Rating'], kde=True)
plt.title("Distribution of Dining Ratings")
plt.show()
```




```
sns.lmplot(
    x='Pricing_for_2',
    y='Dining_Rating',
    data=data,
    height=6
)
plt.title("Price vs Dining Rating")
plt.show()
```




```
sns.pairplot(data[['Dining_Rating','Delivery_Rating','Pricing_for_2']])
plt.show()
```




```
corr = data[['Dining_Rating','Delivery_Rating','Pricing_for_2']].corr()
sns.heatmap(corr, annot=True, cmap='Wistia')
plt.title("Correlation Heatmap")
plt.show()
```